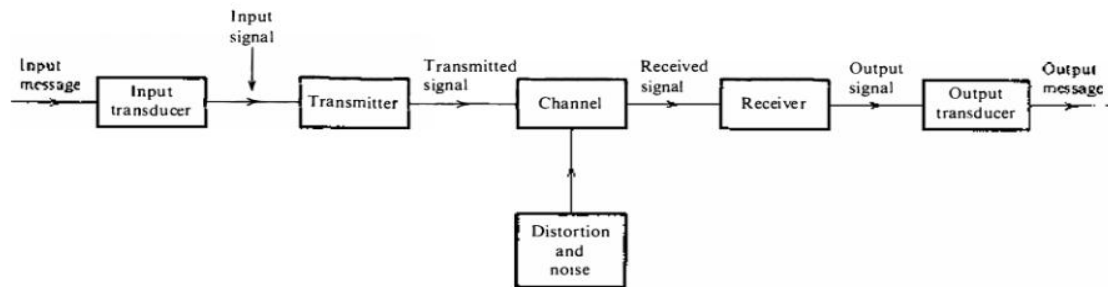


DIGITAL COMMUNICATION SYSTEM



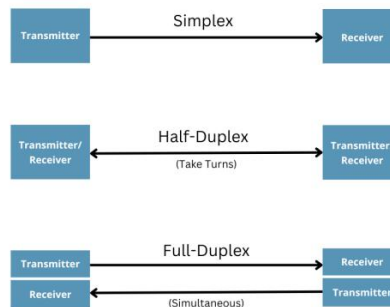
Basic elements of digital communication system:

- **Source**- voice message
- **Input transducer**- physical to electrical
- **Encoder**- error detection, analog to digital
- **Modulator**- make suitable for transmission
- **Medium**- channel like cable, optical fiber
- **Demodulator**- retrieve data from signal
- **Decoder**- remove noise and reconstruct message
- **Output transducer**- electrical signal to sound
- **Destination**- end user

MODES OF COMMUNICATION

Based on data flow:

- **Simplex**- the communication is unidirectional. Example: keyboard, TV, radio
- **Half duplex**- two way but not simultaneous. Example: walkie-talkie
- **Full duplex**- both stations can transmit and receive simultaneously. Example: phone call, video conference

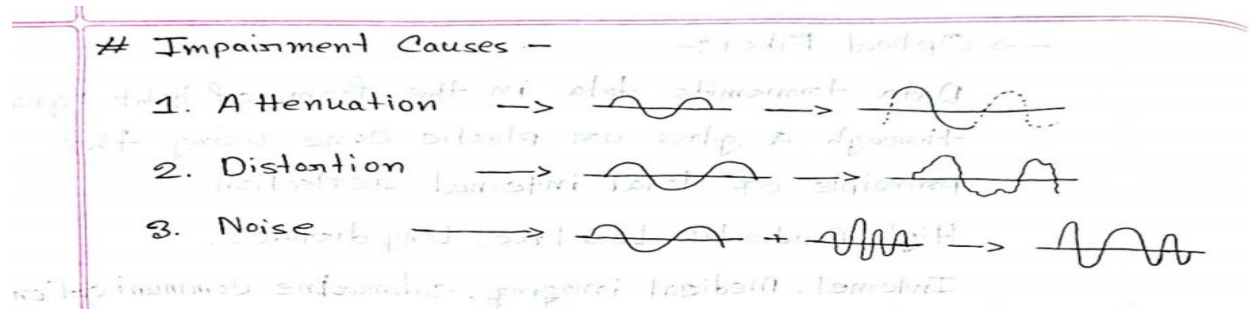


3. Causes of Transmission Impairment

Transmission impairment occurs due to:

1. **Attenuation** – Loss of signal strength over distance.
2. **Distortion** – Change in signal shape due to different frequency speeds.
3. **Noise** – Unwanted signals that interfere with original signal.

These impairments reduce signal quality and cause data errors.



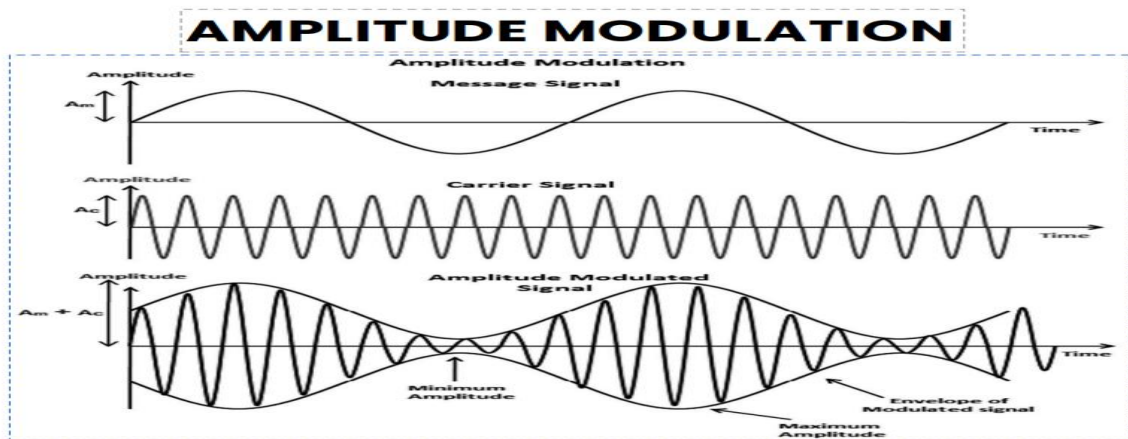
◆ What is Amplitude Modulation?

Amplitude Modulation (AM) is a method where the **amplitude (strength)** of a high-frequency carrier signal is changed according to the **message signal**.

The **frequency and phase** stay the same.

◆ Why do we need AM?

- Low-frequency signals cannot travel far.
- High-frequency signals can travel long distance.
- It allows use of **small antennas**.
- Used in **radio broadcasting**, aviation communication, distress signals, etc.



AMPLITUDE MODULATION

Message/Modulating signal,

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal,

$$c(t) = A_c \cos(2\pi f_c t)$$

By principle of amplitude modulation,

$$\begin{aligned} s(t) &= [A_c + m(t)] \cos(2\pi f_c t) \\ \rightarrow s(t) &= [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t) \\ \rightarrow s(t) &= A_c \left[1 + \frac{A_m}{A_c} \cos(2\pi f_m t) \right] \cos(2\pi f_c t) \\ \rightarrow s(t) &= A_c \cos(2\pi f_c t) + A_c \mu \cos(2\pi f_m t) \cos(2\pi f_c t) \\ \rightarrow s(t) &= A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} \cos 2\pi (f_m + f_c) t + \frac{A_c \mu}{2} \cos 2\pi (f_m - f_c) t \end{aligned}$$

Carrier component
Upper band side
Lower band side

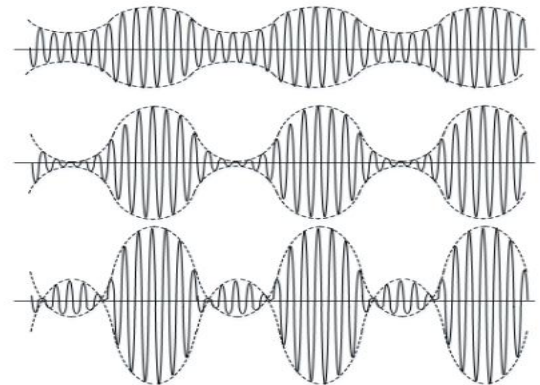
AMPLITUDE MODULATION

Modulating index, $\mu = \frac{A_m}{A_c}$

Under modulation $\mu < 1$

100% modulation $\mu = 1$

Over modulation $\mu > 1$



6.Scenario-Based Questions (AM, FM, PM)

◆ AM (Amplitude Modulation)

Scenario:

A radio station wants long-distance communication using simple receiver.

Answer:

AM is suitable because:

- Simple circuit
 - Cheap receiver
 - Easy demodulation
- But noise affects more.

◆ FM (Frequency Modulation)

In FM, frequency changes according to message signal.

$$f = f_c + \Delta f$$

Scenario:

Music broadcasting where sound quality is important.

Answer:

FM is better because:

- Less noise
- Better sound quality
- Larger bandwidth

◆ PM (Phase Modulation)

In PM, phase changes according to message signal.

$$s(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$

Scenario:

Digital communication system requiring stable amplitude.

Answer:

PM is good because:

- Better noise immunity than AM
- Used in digital systems

🎯 Final Answer (Exam Type)

- For sound quality → FM is best
- For simple & cheap system → AM is best
- For digital communication → PM is best

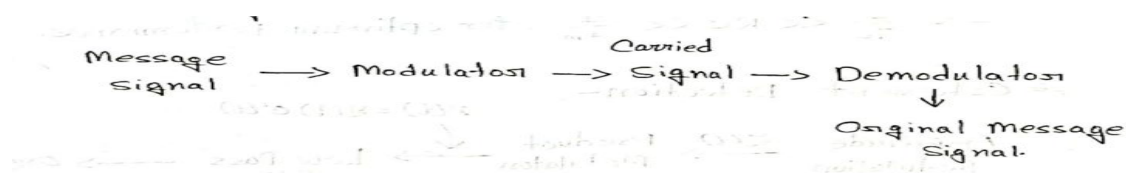
👉 Generally, FM is better than AM, and PM is used more in digital systems.

7. Demodulation – Easy English & Short

Demodulation is the process of **getting back the original message signal** from a modulated signal.

👉 In simple words:

It is the **reverse process of modulation**.



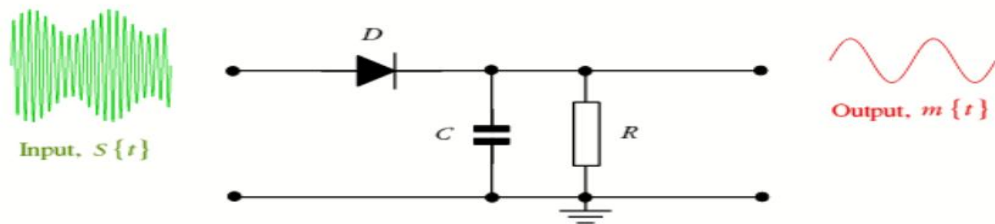
8. Types of amplitude demodulation:

1. Envelope Detector (AM Demodulation)

Envelope Detector is a **simple AM demodulation method** used to recover the original message signal.

ENVELOPE DETECTOR CIRCUIT

- ❖ Works when diode forward biased
- ❖ When voltage increases capacitor stores energy, then discharge when voltage drops
- ❖ Negative cycle is clipped
- ❖ Information loss



2. Coherent / Synchronous Demodulation

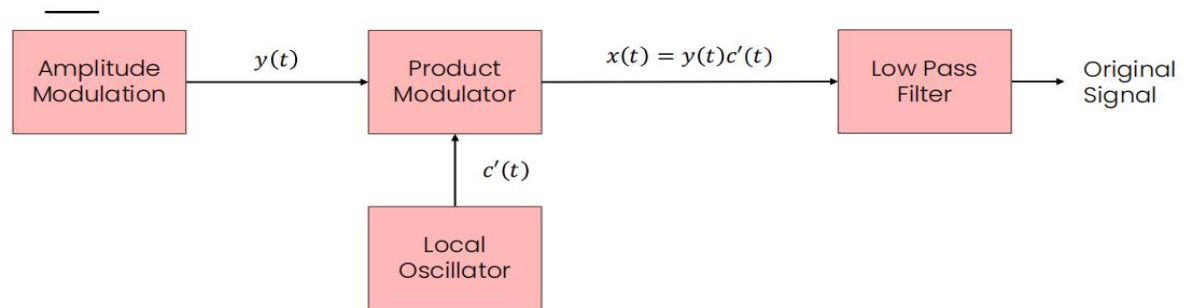
◆ What is it?

Coherent demodulation uses a **locally generated carrier signal** that has the **same frequency and phase** as the transmitted carrier.

◆ Working:

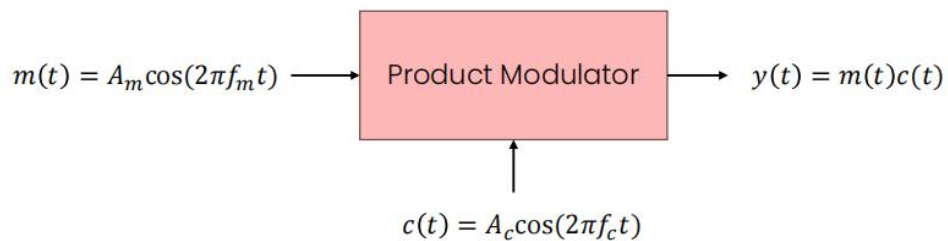
- Multiply received signal with local carrier.
- Pass through low pass filter.
- Original message is recovered.

COHERENT DETECTION



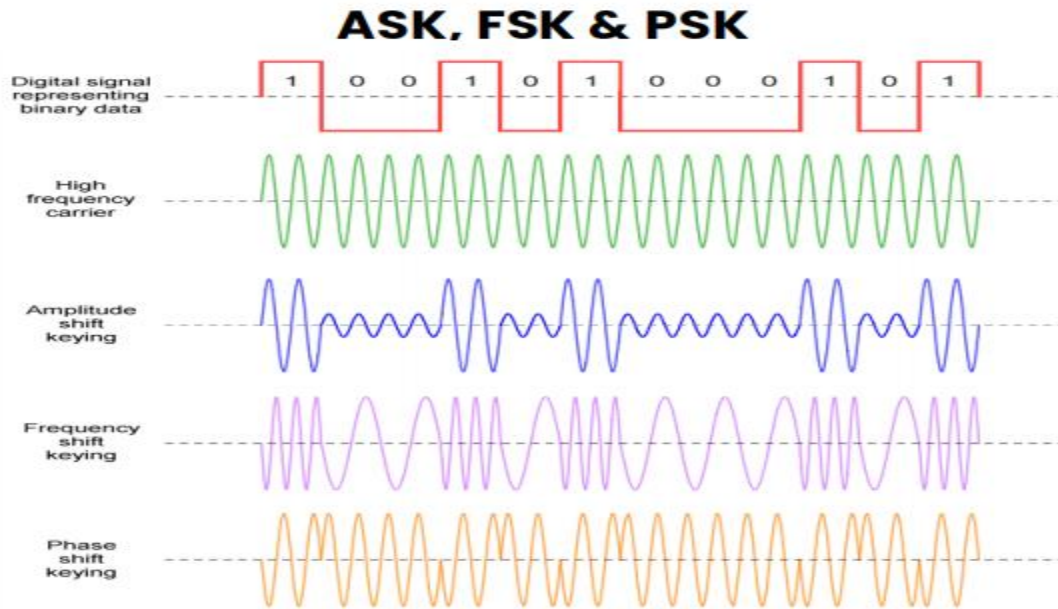
DSBSC AMPLITUDE MODULATION

Double Sideband Suppressed Carrier or DSBSC is AM signal without carrier signal. We only send USB and LSB not the carrier signal for efficiency.



Characteristics:

- No carrier signal
- Sidebands carrying information
- 180 phase reversal at zero crossing of modulating signal
- Same bandwidth as AM
- Complex demodulation



◆ ASK (Amplitude Shift Keying)

- Amplitude changes
- Frequency & phase constant
- Binary 1 → High amplitude
- Binary 0 → Low amplitude

Simple but noise sensitive.

◆ FSK (Frequency Shift Keying)

- Frequency changes
- Amplitude constant
- 1 → High frequency
- 0 → Low frequency

Better than ASK.

◆ PSK (Phase Shift Keying)

- Phase changes
- Amplitude constant

Binary PSK:

- 0 → 0° phase
- 1 → 180° phase

Most reliable among three.

FM (Frequency Modulation)

Advantages:

- Better noise resistance
- Good sound quality
- Less signal distortion

Disadvantages:

- Requires larger bandwidth
- More complex and costly equipment

PM (Phase Modulation)

Advantages:

- Better noise immunity than AM
- Good signal quality
- Efficient transmission

Disadvantages:

- Complex circuit design
- Requires larger bandwidth

In short: FM and PM give better quality than AM but need more bandwidth and complex circuits.